Probability

Year 4 · **Homework Worksheet**

Worked Example

Probability Practice: A bag contains 6 red and 9 blue counters. A counter is picked at random. Find P(red) as a fraction, and P(not red).

Answer: P(red) = 6/15 = 2/5. P(not red) = 1 − 2/5 = 3/5.

Probability Basics (15 questions)

1. A fair six-sided dice is rolled once. What is the probability of rolling a 6? Give your answer as a fraction.
2. A spinner has 5 equal sections numbered 1 to 5. What is the probability of landing on an even number?
3. Write the probability 'impossible' as a number. A) 0 B) 0.25 C) 0.5 D) 0.75 E) 1
4. A bag contains 20 balls, 5 of which are green. What is the probability of picking a green ball, as a decimal?
5. Order these words from least likely to most likely: certain, unlikely, impossible, likely, evens.
6. A coin is flipped once. What is the probability of landing on heads, as a percentage?
7. Which word best describes an event with probability 0.9? A) Impossible B) Unlikely C) Evens D) Likely E) Certain
8. A bag contains only white counters. A counter is picked at random. What is the probability it is black?
9. List all the possible outcomes when a spinner with sections 1, 2 and 3 is spun once.
10. A box contains 9 green pens and 1 orange pen. What is the probability of picking an orange pen, as a fraction?
11. Select ALL that apply: which of these could be the probability of an event? A) 1 B) −0.5 C) 0.75 D) 2 E) 0
12. A fair six-sided dice is rolled once. What is the probability of rolling a number less than 3? A) 1/6 B) 1/3 C) 1/2 D) 2/3 E) 5/6
13. The probability that a bus is on time is 17/20. Write this probability as a percentage.
14. List all the possible outcomes when a coin is flipped and then a second coin is flipped.
15. A spinner is divided into 5 equal sections coloured red, red, blue, green and yellow. What is the probability of landing on red?

Applying Probability (15 questions)

1. A bag contains 6 blue, 2 red and 2 yellow counters. A counter is picked at random. What is the probability it is NOT blue? Give your answer as a fraction.
2. A spinner has 8 equal sections: 5 are white, 3 are black. What is the probability of landing on black, as a decimal?
3. The probability that a football team wins a match is 0.55. What is the probability that they do not win? A) 0.35 B) 0.4 C) 0.45 D) 0.5 E) 0.55
4. A fair coin is flipped twice. List all 4 possible outcomes, then state the probability of getting at least one tail.
5. A jar contains 25 marbles: 10 are striped, the rest are plain. What is the probability of picking a plain marble, as a percentage?
6. Two fair coins are flipped. What is the probability of getting two heads or two tails (not a mix)? A) 1/4 B) 1/2 C) 3/4 D) 1 E) 0

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7. A bag has 5 orange and 3 purple counters. A counter is picked, its colour noted, and it is NOT put back. If the first counter picked is purple, what is the probability the second counter picked is also purple?

8. Select ALL that apply: a fair spinner numbered 1 to 8 is spun once. Which statements are true? A) P(spinning a number greater than 8) = 0. B) P(spinning an odd number) = 1/2. C) P(spinning a number less than 9) = 1. D) P(spinning an 8) = 1/8. E) P(spinning a multiple of 5) = 1/4.

9. A box of 20 biscuits has 12 plain and 8 chocolate biscuits. What is the probability of picking a chocolate biscuit, as a fraction in its simplest form?

10. A weather forecaster says there is an 80% chance of rain tomorrow. What is the probability, as a fraction, that it will NOT rain?

11. A bag contains only green and yellow counters. P(green) = 0.45. If there are 20 counters in total, how many are yellow?

12. A fair spinner has sections numbered 1 to 4. It is spun twice. What is the probability that it lands on 1 both times? A) 1/16 B) 1/8 C) 1/4 D) 1/2 E) 2/16

13. A drawer contains 5 red gloves and 5 grey gloves. One glove is pulled out at random, not put back, and a second glove is pulled out. What is the probability that both gloves are red?

14. In a raffle, 150 tickets are sold and Tom buys 3 of them. What is the probability that Tom's ticket is drawn as the winner, as a decimal?

15. Select ALL that apply: a bag contains 2 red, 5 blue and 3 green counters (10 total). Which statements are true? A) P(red) = 1/5. B) P(not blue) = 1/2. C) P(green) = 0.3. D) P(blue) = 0.5. E) P(red or green) = 0.6.

Standard / Advanced (15 questions)

1. A bag always has the same mix of red and total counters: 2 out of every 6 counters are red. If a bigger bag has 30 counters with the same mix, how many are red?

2. If you flip a coin 3 times in a row, how many different sequences of heads and tails could you get in total? A) 4 B) 6 C) 8 D) 10 E) 12

3. Numbers 1 through 9 are marked in equal sections around a spinner. What is the probability that a single spin lands on a multiple of 3?

4. A jar contains 3 green marbles and 1 red marble. What is the probability of picking the red marble, written as a percentage, then that percentage doubled?

5. When two fair dice are rolled and the scores added together, one particular total comes up more often than any other. Which total is it, and what is the probability of rolling it?

6. Counters sit in a box in the ratio 1:1:2 for red, blue and green. Pick one counter without looking. What is the probability that it is anything other than red?

7. Among the counters in a bag are 4 black ones and 6 white ones. Two counters are drawn one after the other, the first not going back in. What is the probability both turn out white? A) 1/3 B) 3/10 C) 2/5 D) 7/22 E) 1/2

8. There's a 50% forecast chance of rain on Saturday, and separately a 20% chance on Sunday, with one day's weather not affecting the other. Find the probability that both days see rain.

9. Only red and blue counters sit in a bag. Red counters make up 1/4 of the total, and there happen to be 18 blue ones. How many counters are in the bag altogether? A) 20 B) 22 C) 24 D) 26 E) 28

10. In a group of 40 students, the probability a randomly chosen student studies Art is 1/2. Of those who study Art, 1/4 also study Drama. What is the probability that a randomly chosen student studies both Art and Drama?

11. Spin a spinner labelled 1 to 4 twice in a row, adding the two numbers you get. What is the probability that this sum is odd?

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12. Select ALL that apply: a bag contains 3 red and 7 blue counters. Two counters are picked one after another without replacement. Which statements are true? A) P(first is red) = 3/10. B) P(both red) = 1/15. C) P(first red, second blue) = 7/30. D) P(both blue) = 7/15. E) P(both red) = P(both blue).

13. Roll a standard six-sided dice two times and multiply the two scores together. What is the probability that this product comes out above 25?

14. A bag holds 15 counters, a mix of red and blue. Over many repeated picks (each counter replaced after being noted), red comes up twice as often as blue on average. How many red counters are in the bag?

15. Six tickets, numbered 1 to 6, sit in a box. Two are drawn out together at the same time (the order they come out in doesn't matter). What is the probability that both tickets show even numbers?

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